

Professional Practices in IT (CS 4001)

Date: 23rd September, 2025

Course Instructor(s):

Mr. M. Waqas, Mr. Zummar, Mr. Waqar, Mr.
Amjad, Mr. Zeyaad, Mr. Amir, Mr. Saifullah

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 36

Total Questions: 4

Roll No

Section

Student Signature

Please answer all questions except question2 on answer sheet in the same order as they appear on question paper. Answer question2 on the question paper and attach question paper with answer sheet. Each part/task answer should not be more than 3 lines. Answer of each task/part more than 3 lines will not be considered. Write to the point.

Following Instructions are ONLY for only BDS-7C and BSSE-7C:

Question 2 is a bonus question for the sections BDS-7C and BSSE-7C. However, the students of these sections should note that they MUST attempt the other three questions first. Bonus marks shall be considered only if they have properly attempted the rest of the three (3) questions. Maximum marks will not exceed 100%. They may choose not to attempt it as well.

CLO 2: Demonstrate familiarity with an existing code of ethics in software engineering and apply it to simple case studies.

Question1: A prominent AI ethics researcher, Dr. Anya Sharma, stated in a recent conference that "the AI developer and data scientist of today operate in a professional vacuum, lacking the formal governance and ethical accountability of a licensed engineer."

Discuss this statement by addressing the following points in your answer: [4x2]

Part1. Professional Standards and Licensing: Compare the process of becoming a licensed engineer with the current lack of a formal licensing process for AI developers. Explain why this difference is significant.

The process of becoming a **licensed engineer** is a rigorous, legally defined pathway, typically involving: a specific accredited degree, several years of supervised work experience, and passing one or more comprehensive professional exams (e.g., the Fundamentals of Engineering and Principles and Practice of Engineering exams). Licensure provides a legal right to practice and confirms a **baseline standard of competency** and adherence to public safety.

In contrast, there is currently **no formal licensing process for AI developers or data scientists**. Anyone can claim the title based on self-taught skills, bootcamps, or non-accredited degrees.

This difference is **significant** because licensing ensures **public protection**. Without it, there is no standardized assurance that an AI professional building systems that affect public safety, health, or

National University of Computer and Emerging Sciences

Lahore Campus

financial well-being (e.g., medical diagnostics, self-driving cars) has met a minimum threshold of knowledge or ethical training enforceable by law.

Part2. Ethical Codes and Enforcement: Contrast the formal, enforced ethical codes of engineering professional societies with the informal, often self-regulated ethical guidelines in AI development. Provide an example of how this difference could lead to a negative outcome in a real-world scenario.

Engineering ethics are formal, mandatory, and enforced by professional bodies (like the National Society of Professional Engineers) and government licensing boards. Violating these codes can result in **losing one's license** and the ability to practice, which is a powerful deterrent. The core duty is the protection of the health, safety, and welfare of the public.

AI ethics are largely **informal, voluntary, and self-regulated**. They exist primarily as corporate guidelines, academic principles, or ethical manifestos (like the ACM Code of Ethics) but lack a regulatory body with the power to bar an individual from working.

Real-world Negative Outcome Example:

- **Scenario:** An AI developer working for a credit agency notices their algorithm is systematically denying loans to applicants from a specific low-income zip code, despite the applicants having similar credit profiles to approved individuals from wealthier areas (algorithmic bias).
- **Outcome:** Because there is no external, enforced ethical code or licensing board, the developer's primary obligation is to their employer. If the employer deems the bias acceptable (or too expensive to fix), the developer may continue deploying the biased system without professional consequence, perpetuating systemic economic harm. An engineer, facing a similar ethical issue concerning public safety, would be obligated by their license to report or fix the flaw, or risk losing their entire career.

Part3. Accountability and Liability: Using a concrete example for each, explain how professional accountability and liability differ between a civil engineer whose project fails and an AI developer whose algorithm causes harm.

National University of Computer and Emerging Sciences

Lahore Campus

Part 3. Accountability and Liability

Profession	Concrete Example of Failure	Accountability and Liability
Civil Engineer	A bridge designer cuts corners on structural testing, and the bridge collapses, resulting in injuries.	The engineer faces professional accountability (loss of license), civil liability (lawsuits for damages), and potentially criminal liability (manslaughter or negligence) because the fault is directly traceable to their licensed design decisions.
AI Developer	An AI-driven hiring algorithm rejects all female candidates for a technical role due to training data bias, resulting in a discrimination lawsuit against the company.	Accountability primarily falls on the company (civil liability/fines). The individual AI developer who trained or deployed the model is rarely held personally or professionally liable. The complexity and opacity of the algorithm (the "black box" problem) make it difficult to trace the harmful outcome back to a specific individual's error or negligence.

The key difference is that the **engineer is held accountable to the public and the law** via their license, while the **AI developer is primarily accountable to their employer**.

Part4. The Future of Professional Responsibility: Based on your analysis, discuss what steps could be taken to professionalize the field of AI development to align it more closely with the engineering profession.

To professionalize AI development and align it more closely with the engineering profession, the following steps could be taken:

1. **Mandatory Certification and Continuing Education:** Establish a non-governmental professional body (similar to the American Medical Association or engineering boards) that grants a "Certified AI Professional" status based on passing technical and ethical exams. This certification would require mandatory continuing education (e.g., on fairness and security updates) to remain valid.
2. **Define a Fiduciary Duty:** Create legal or industry standards that impose a **fiduciary duty** (a legal obligation to act in the best interest of the client or the public) on AI developers working on high-impact systems (e.g., health, finance, criminal justice). This would legally elevate their responsibility above merely following employer directives.
3. **Algorithmic Accountability Audits:** Mandate that high-risk AI systems (e.g., those used by government or financial institutions) undergo independent, third-party **ethical and bias audits** conducted by certified professionals, similar to how financial audits or structural inspections are mandatory.
4. **Specialized Ethics and Law Training:** Integrate required, accredited coursework in **AI Ethics, Data Law, and Bias Mitigation** into all university and major boot camp programs that train AI professionals, making these topics non-negotiable foundations of the curriculum.

National University of Computer and Emerging Sciences

Lahore Campus

CLO 2: Demonstrate familiarity with an existing code of ethics in software engineering and apply it to simple case studies.

Question2: Identify each of the following clause as either from CODE OF PRACTICE or CODE OF CONDUCT. [10]

Code of conduct; standards of behaviors of members

Code of practice; best/standard way to practice your profession

Clause	Answer
All staff must follow national data protection laws (e.g., GDPR, HIPAA) and organizational compliance rules	COC
All systems should undergo unit, integration, and user-acceptance testing before deployment to minimize risk.	COP
Applications should be designed in compliance with accessibility standards (e.g., WCAG) to support all users, including those with disabilities.	COP
Company IT systems, networks, and software must not be used for personal profit or unlawful activities.	COC
Developers should follow secure coding practices to prevent vulnerabilities such as SQL injection or buffer overflow exploits.	COP
Employees must declare any external business or personal interest that could compromise their professional decisions.	COC
Staff must avoid harassment, discrimination, or offensive behavior in all professional interactions.	COC
Organizations should keep user data only as long as necessary for operational or legal requirements, then securely delete it.	COP
Software developers should maintain clear and complete project documentation to ensure maintainability and knowledge transfer.	COP
Employees must not share sensitive company or client information with unauthorized individuals.	COC

.CLO 4: Articulate and reflect on the industry expectations of competence and conduct in computing professions.

National University of Computer and Emerging Sciences

Lahore Campus

Question3: The "Eco-Logistics" Venture

Background

Three friends—Aisha, Ben, and Chloe—met during their Master's program in Supply Chain Management and developed an innovative idea for a software solution. Their concept, named "Eco-Logistics," is a cloud-based platform that helps small to medium-sized delivery companies optimize their routes to reduce fuel consumption and carbon emissions. The platform uses a proprietary algorithm developed by the team.

- **Aisha** is a software engineer with over five years of experience. She is the lead developer of the core algorithm and the mobile app. She has invested \$10,000 of her personal savings into the project.
- **Ben** is a business development expert with a strong network in the logistics industry. He has no upfront capital to invest but has secured a potential pilot project with a local courier company, which, if successful, could provide their first significant revenue stream.
- **Chloe** is a finance and marketing professional. She has a high net worth and is willing to invest \$50,000 to cover initial operational costs, server fees, and marketing. She insists on a clear structure to protect her investment and ensure sound financial management.

The friends are excited about the potential of Eco-Logistics. They have a working prototype and are ready to take the next step: formalizing their business structure. They are committed to the business but have different priorities and levels of risk tolerance. Aisha is concerned about her personal liability, Ben is focused on a flexible, fast-paced launch, and Chloe wants her financial contributions and influence to be legally protected.

The pilot project is scheduled for six months from now, and they anticipate needing to hire at least two more developers and a sales manager within the next year to expand their user base.

Tasks

Based on the "Eco-Logistics" case study, answer the following questions: [3+2+5]

1. **Propose and Compare the legal implications of forming a partnership versus a private limited company.** Your answer must recommend which structure is more suitable for this specific situation, considering the founders' individual priorities.

1. Legal Implications: Partnership vs. Private Limited Company

Based on the "Eco-Logistics" case study, the founders must choose between a **partnership** and a **private limited company**. While a partnership is simpler to form, a private limited company offers significant legal and financial advantages that make it the more suitable choice for this venture.

Aspect	Partnership	Private Limited Company (Pvt. Ltd.)
Legal Status	Not a separate legal entity. The partners are the business.	A separate legal entity from its owners (shareholders).
Liability	Unlimited liability. Partners are personally responsible for all business debts. If the business fails, personal assets (e.g., homes, savings) can be at risk.	Limited liability. The owners' liability is limited to the capital they invested. Personal assets are protected from business debts.
Capital & Ownership	Capital contributions can be informal. Ownership is typically shared equally	Ownership is divided into shares. The company can sell shares to raise capital

National University of Computer and Emerging Sciences

Lahore Campus

	unless a partnership agreement states otherwise.	from investors without the need for a loan.
Management & Control	All partners have an equal right to manage the business, which can lead to disputes if roles and responsibilities are not clearly defined.	The company is managed by directors, who can be different from the shareholders. This allows for a clear division between ownership and management.
Continuity	The partnership dissolves if a partner leaves or dies.	The company has perpetual succession . It continues to exist regardless of changes in ownership or management.

Recommendation: A **private limited company** is the most suitable structure for Eco-Logistics. The primary reason is **limited liability**, which is crucial for protecting Chloe's significant investment (\$50,000) and Aisha's personal assets. The limited company structure also makes it easier to raise capital in the future by selling shares, which aligns with their growth plans. Furthermore, it provides the formal governance Chloe desires, with clear rules for management and decision-making, which can prevent future disputes between the founders.

2. *Recommend an organizational model (bureaucratic, organic, or matrix) that would be most suitable for the "Eco-Logistics" team as it grows. Justify your recommendation by explaining how the chosen model will support their business goals and future growth, while also addressing the needs of the individual founders.*

For "Eco-Logistics" one year from now, an **organic organizational model** at a small scale, is most suitable. This model balances the need for innovation in development.

3. Based on the "Eco-Logistics" case study, assume the company has successfully completed its pilot project and is now seeking to expand. The team anticipates hiring at least two more developers and a sales manager within the next year.

Propose a specific organizational chart for Eco-Logistics one year from now. Justify your proposed structure by explaining how it:

- **Balances the need for ongoing innovation with the demands of a growing business**

CLO 4: Articulate and reflect on the industry expectations of competence and conduct in computing professions.

Question4: TechNova Innovations Ltd. is a rapidly growing startup specializing in developing sustainable AI solutions. The company's founders, having relied on a mix of personal savings and angel investments for the first three years, are now planning a major expansion. Their goal is to build a new research facility and scale up their marketing efforts to enter international markets. To finance this expansion, the board of directors is considering two primary options:

Option A: Debt Financing

- Secure a 10-year loan of **\$5 million** from a commercial bank. The loan has an annual interest rate of **7%**.

National University of Computer and Emerging Sciences

Lahore Campus

- The board is confident they can meet the interest payments, as the company's current and projected revenues are strong. However, some board members are concerned about the financial risk this would introduce, particularly if there's a market downturn.

Option B: Equity Financing

- Issue new shares to a venture capital firm, raising **\$5 million**.
- This option would not incur any interest payments but would dilute the ownership of the original founders and existing shareholders. The new investors would also demand a seat on the board and a say in the company's strategic direction.

TechNova's financial records for the previous year are as follows:

- **Total Equity:** \$10 million
- **Total Assets:** \$12 million
- **Total Liabilities:** \$2 million (consisting of operational debts and existing loans)
- **Net Income:** \$1.5 million
- **EBIT (Earnings Before Interest and Taxes):** \$2.5 million

The board needs to make a decision on how to proceed.

Tasks

Based on the case study of TechNova Innovations Ltd., answer the following questions: [2+6]

1. **Calculate the current gearing ratio** of TechNova Innovations Ltd. using the Debt-to-Equity formula. Clearly show your calculation.
2. **Objectively analyze the gearing situation** for TechNova if they choose **Option A** (Debt Financing). Use your calculated ratio and other financial data from the case study to explain the implications of this decision. Discuss both the **benefits** and the **risks** associated with increasing the company's financial leverage.

1. Calculation of the Current Gearing Ratio

The gearing ratio measures the proportion of a company's capital that comes from debt. The Debt-to-Equity formula is used to calculate this, as shown below:

Gearing Ratio = Total Equity / Total Liabilities

Using the provided financial data for TechNova Innovations Ltd.:

- **Total Liabilities:** \$2 million
- **Total Equity:** \$10 million

$$\text{Gearing Ratio} = \frac{\$2 \text{ million}}{\$10 \text{ million}} = 0.2 \text{ or } 20\%$$

TechNova's current gearing ratio is **20%**. This indicates that 20% of the company's capital structure is financed by debt, with the remaining 80% financed by equity.

2. Analysis of the Gearing Situation after Debt Financing

National University of Computer and Emerging Sciences

Lahore Campus

If TechNova chooses **Option A** (Debt Financing), the company's gearing situation will change significantly. The current \$2 million in liabilities would increase by the new \$5 million loan, bringing the new total liabilities to **\$7 million**. Equity would remain unchanged at \$10 million.

The new gearing ratio would be:

$$\text{Gearing Ratio} = \frac{\$7 \text{ million}}{\$10 \text{ million}} = 0.7 \text{ or } 70\%$$

Benefits (Leverage)

Increasing leverage allows a company to grow using external funds without diluting ownership. For TechNova, the key benefits are:

- **Increased Returns to Shareholders:** By using debt to finance growth, the company can potentially generate higher profits that benefit the existing shareholders. As long as the return on investment from the new facility exceeds the 7% interest rate on the loan, the company's profitability and earnings per share will increase.
- **Tax Benefits:** Interest payments on debt are typically tax-deductible, which reduces the company's overall tax liability.

Risks (Financial Risk)

A higher gearing ratio makes the company more vulnerable to financial risk. The key risks for TechNova are:

- **Increased Financial Burden:** The company will be obligated to make fixed annual interest payments of \$350,000 (\$5 million * 7%), regardless of its profitability. If a market downturn occurs and revenue drops, meeting these payments could become difficult, potentially leading to financial distress or even bankruptcy.
- **Reduced Flexibility:** A high debt load can make it difficult for TechNova to secure additional financing in the future, as lenders may view the company as too risky. This could limit its ability to respond to future opportunities or unforeseen challenges.

In summary, choosing debt financing would provide TechNova with the capital to expand without diluting ownership, but it would also expose the company to a significantly higher level of financial risk due to its increased leverage.